

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2388

Cross 138 kV
May 26, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2388. The Interconnection Customer proposes a 116.8 MW solar plus storage facility interconnecting at the Cross 138 kV substation in the West Virginia area. The request was received on March 03, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 471 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,163,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: E. Ripley to W Fork 138 kV line
Contingency: West Woodland to Menlo 138 kV
Loading: 149.9% (Rating: 236 MVA)
DFAX: 5.9%
Pre-project loading: 146.9%
2. Facility: Carlsbad Waterfield to Menlo Substation 138 kV
Contingency: Menlo to West Woodland 138 kV
Loading: 142.2% (Rating: 261 MVA)
DFAX: 5.9%
Pre-project loading: 139.5%
3. Facility: Carlsbad Waterfield to E. Ripley Substation 138 kV
Contingency: Menlo to W Woodland 138 kV
Loading: 131.0% (Rating: 277 MVA)
DFAX: 5.9%
Pre-project loading: 128.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E. Ripley to W Fork 138 kV line	Rebuild line with higher-capacity conductor	\$5,616,000	14% (\$797,000)
Carlsbad Waterfield to Menlo Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,233,000	14% (\$459,000)
E. Ripley to Carlsbad Waterfield 138 kV	Rebuild line with higher-capacity conductor	\$6,387,000	14% (\$907,000)
	Total allocated to project		\$2,163,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.